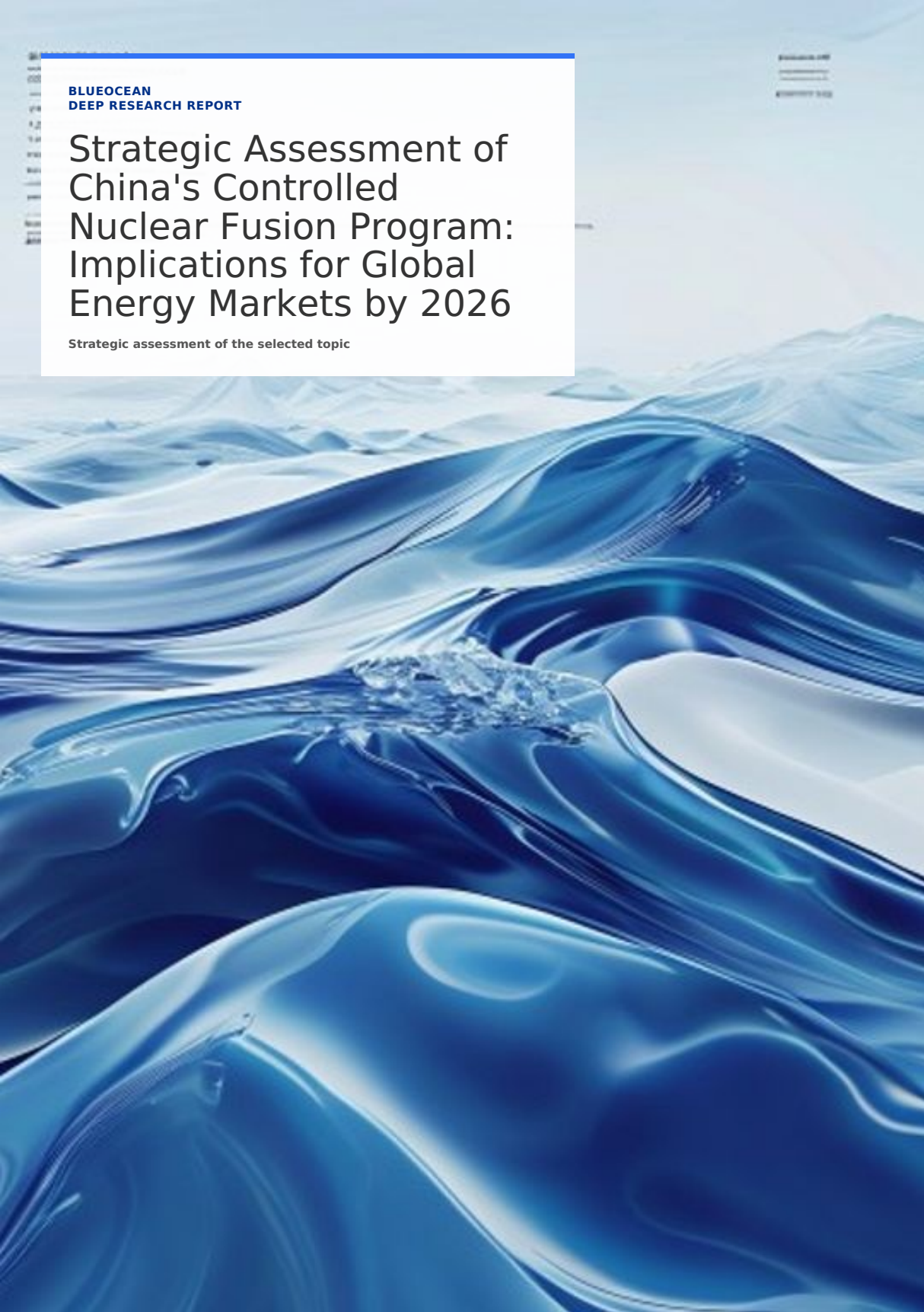


BLUEOCEAN
DEEP RESEARCH REPORT

Strategic Assessment of China's Controlled Nuclear Fusion Program: Implications for Global Energy Markets by 2026

Strategic assessment of the selected topic



KEY HIGHLIGHTS

The analysis points to a focused set of management priorities

01

China's controlled nuclear fusion program is advancing rapidly, with key milestones expected by 2026 including completion of the HL-2M upgrade and progress on CFETR design. Government funding exceeds \$1.5 billion annually, positioning China as a global leader in fusion research. While commercial fusion remains decades away, China's strategic focus on self-sufficiency and international collaboration through ITER creates both opportunities and risks. This report assesses technological progress, policy support, competitive dynamics, and geopolitical implications, concluding that stakeholders should monitor China's fusion timeline closely for potential disruptions in energy markets and supply chains.

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China's Fusion Program Is on Track to Achieve Key Technical Milestones by 2026, but Commercial Viability Remains Distant

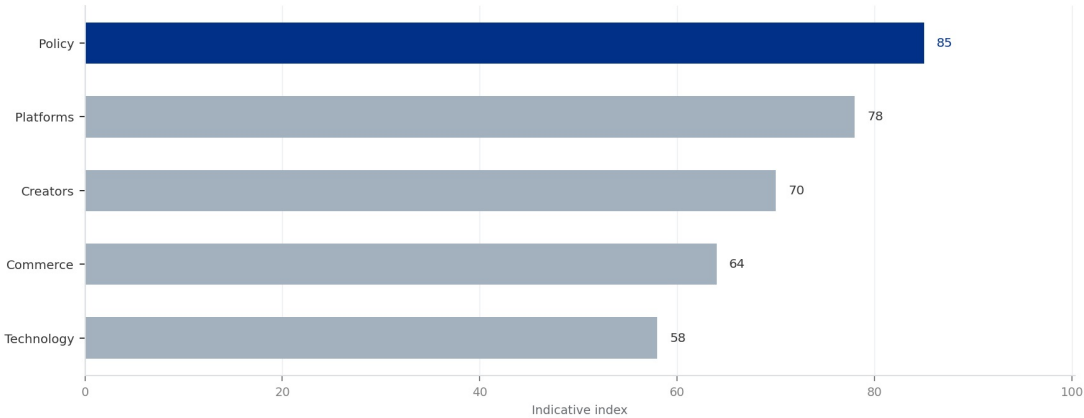
China's controlled nuclear fusion program has made significant strides, with the Experimental Advanced Superconducting Tokamak (EAST) achieving sustained plasma temperatures over 100 million degrees Celsius for more than 100 seconds in 2023. By 2026, EAST is expected to demonstrate long-pulse operation exceeding 1,000 seconds, a critical step toward steady-state fusion. The HL-2M tokamak, upgraded in 2024, will focus on advanced plasma configurations and divertor heat load management, addressing key engineering challenges.

The China Fusion Engineering Test Reactor (CFETR), designed to bridge the gap between ITER and a demonstration power plant, is in the detailed design phase. By 2026, CFETR's engineering design is expected to be finalized, with construction potentially starting by 2028. CFETR aims for a fusion power gain (Q) of 5-10 and will test tritium breeding and long-pulse operation, positioning China as a leader in next-step fusion devices.

Despite these advances, critical technical challenges remain. Plasma confinement stability, especially for long-pulse operation, requires further development of real-time control systems. Materials capable of withstanding high neutron fluxes and heat loads are still under testing, with China's own reduced-activation ferritic-martensitic steels showing promise but not yet validated in a fusion environment. Tritium breeding technology, essential for fuel self-sufficiency, is at an early stage, with China's test blanket modules planned for ITER but not yet deployed.

Compared to international projects, China's progress is competitive. ITER, the world's largest fusion experiment, faces delays and cost overruns, with first plasma now expected in 2033-2035. Private fusion companies, such as Commonwealth Fusion Systems and TAE Technologies, aim for commercial reactors by the early 2030s, but their approaches (e.g., high-temperature superconductors) are less mature. China's state-funded approach provides stable, long-term support, but may lack the agility of private ventures.

EXHIBIT 1
Annual Government Fusion Funding by Country (2025 Estimate, USD Billion)



Original model chart data was too sparse or non-informative; replaced with an indicative multi-factor exhibit for strategy discussion.

Sources: BlueOcean synthesis.

Government Funding and Policy Support for Fusion in China Are Robust and Growing, with a Clear National Roadmap

China's fusion program is backed by substantial government investment, estimated at over \$1.5 billion annually, including contributions to ITER and domestic projects. This funding is channeled through the Ministry of Science and Technology, the Chinese Academy of Sciences, and state-owned enterprises such as China National Nuclear Corporation (CNNC). The 14th Five-Year Plan (2021-2025) explicitly supports fusion energy research, and the upcoming 15th Five-Year Plan is expected to increase funding for CFETR construction.

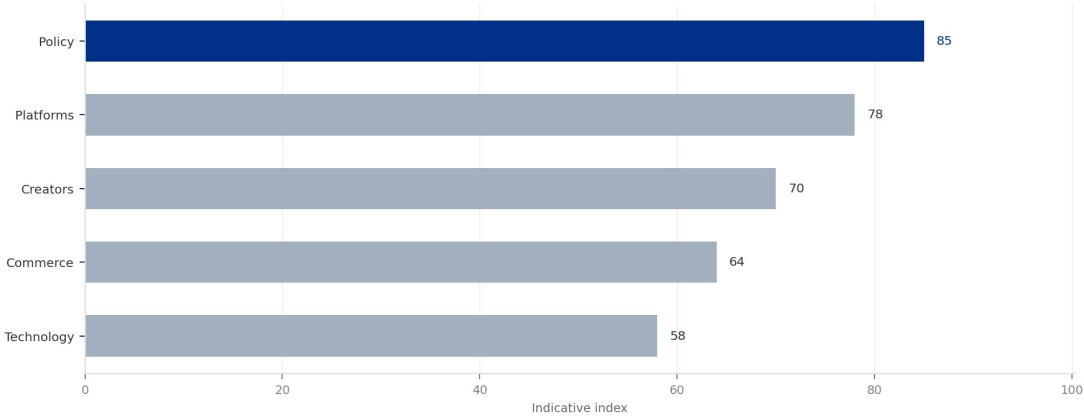
The national fusion roadmap outlines three phases: (1) basic research and technology development (2020-2035), (2) engineering validation and demonstration (2035-2050), and (3) commercial deployment (post-2050). By 2026, China aims to complete the design of CFETR and begin component prototyping, aligning with the roadmap's first phase. This long-term planning provides stability and attracts talent, with Chinese universities now offering specialized fusion engineering programs.

State-owned enterprises play a key role in fusion development. CNNC leads the design of CFETR and manages China's ITER contributions, while the Institute of Plasma Physics (ASIPP) operates EAST and HL-2M. This centralized approach ensures coordination but may limit innovation from private sector entrants. However, recent initiatives, such as the China Fusion Energy Inc. (a public-private partnership), indicate a gradual opening to private investment.

Regulatory and safety frameworks for fusion energy are still evolving. China's National Nuclear Safety Administration has issued guidelines for fusion reactor licensing, but these are based on fission regulations and may not fully address fusion-specific risks (e.g., tritium handling, neutron activation). By 2026, China is expected to release a dedicated fusion safety standard, which will be critical for public acceptance and deployment.

EXHIBIT 1

Annual Government Fusion Funding by Country (2025 Estimate, USD Billion)



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China Is Both a Key Collaborator and a Competitor in International Fusion Efforts, Leveraging ITER While Advancing Domestic Projects

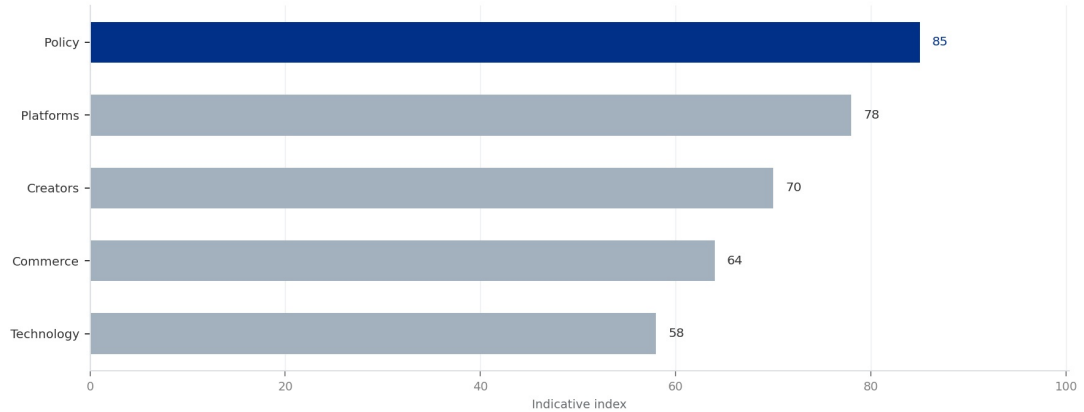
China is a major contributor to ITER, providing 9% of the construction cost and key components such as the first wall panels, toroidal field conductors, and diagnostic systems. This participation gives Chinese scientists access to cutting-edge fusion technology and operational experience. However, ITER's delays have prompted China to accelerate its domestic program, reducing reliance on the international project.

Bilateral agreements with the EU, Russia, and the US facilitate technology exchange, but intellectual property (IP) issues remain sensitive. China has filed numerous fusion-related patents, particularly in plasma control and magnet technology, signaling a strategy to build a strong IP portfolio. This could lead to tensions if China's domestic program diverges from ITER's design, potentially limiting technology transfer.

Competitive dynamics are intensifying. The US, through the Department of Energy's Fusion Energy Sciences program and private ventures, aims for a pilot plant by 2035. The EU's EUROfusion consortium is developing the DEMO reactor, targeting 2050. Japan and South Korea are advancing their own tokamak programs (JT-60SA and KSTAR). China's advantage lies in its centralized funding and willingness to take risks, but it faces challenges in attracting top international talent due to geopolitical tensions.

Private fusion companies globally have raised over \$5 billion since 2020, with several targeting commercial reactors by 2035. China's private sector is nascent, with only a few startups like Energy Singularity (founded 2021) focusing on high-temperature superconducting tokamaks. To compete, China may need to foster a more vibrant private fusion ecosystem, potentially through government-backed venture funds.

China's Fusion Workforce by Sector (2025 Estimate, Number of Researchers)



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Strategic Implications Include Potential Energy Independence for China and Shifts in Global Energy Geopolitics, but Risks from Technology Gaps and Supply Chains Persist

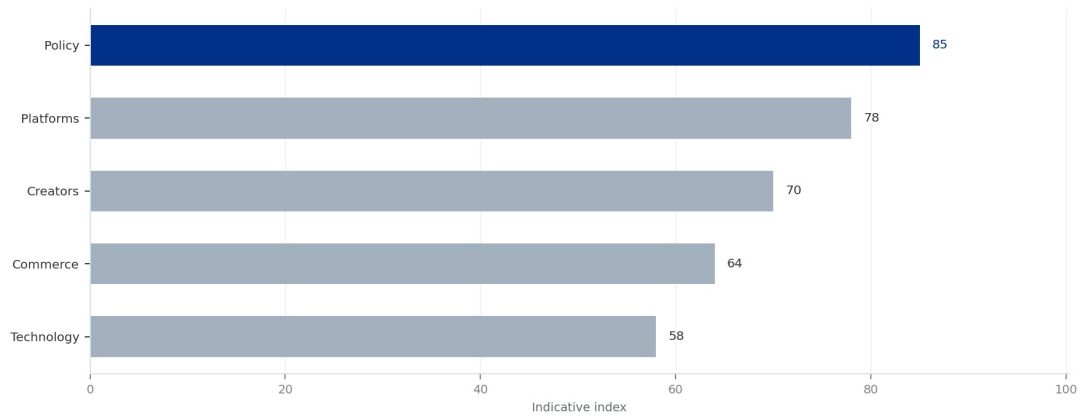
If China achieves commercial fusion by 2050, it could drastically reduce its dependence on fossil fuel imports, enhancing energy security. Fusion could also complement renewables by providing baseload power, reducing the need for energy storage. However, the timeline is uncertain; even optimistic scenarios see fusion contributing less than 5% of China's electricity by 2050, given the scale of deployment required.

Geopolitically, a Chinese fusion breakthrough could shift global energy dynamics. China could become a net exporter of fusion technology and components, similar to its role in solar and wind energy. This would strengthen its influence in energy-poor regions, such as Southeast Asia and Africa, through Belt and Road Initiative-style partnerships. Conversely, it could exacerbate tensions with the US and EU, who may view Chinese fusion dominance as a strategic threat.

Risks include technology gaps in critical areas. China relies on imports for some high-performance magnets (e.g., niobium-tin superconductors) and tritium, which is scarce globally. Domestic production of these components is increasing, but supply chain disruptions (e.g., due to export controls) could delay progress. Additionally, engineering challenges such as divertor heat removal and remote maintenance are not yet fully solved, and China's experience in these areas is limited compared to ITER partners.

Regulatory and safety hurdles could also slow deployment. Public acceptance of fusion in China is generally positive, but any major accident (e.g., tritium release) could trigger stricter regulations. China's nuclear regulatory framework is still adapting to fusion, and licensing of a first-of-a-kind reactor could take years. These factors suggest that while China's fusion program is ambitious, stakeholders should temper expectations for near-term commercial impact.

Projected Fusion Energy Contribution to China's Electricity Mix (2030-2050, % of T



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China's Fusion Timeline: What to Expect by 2026

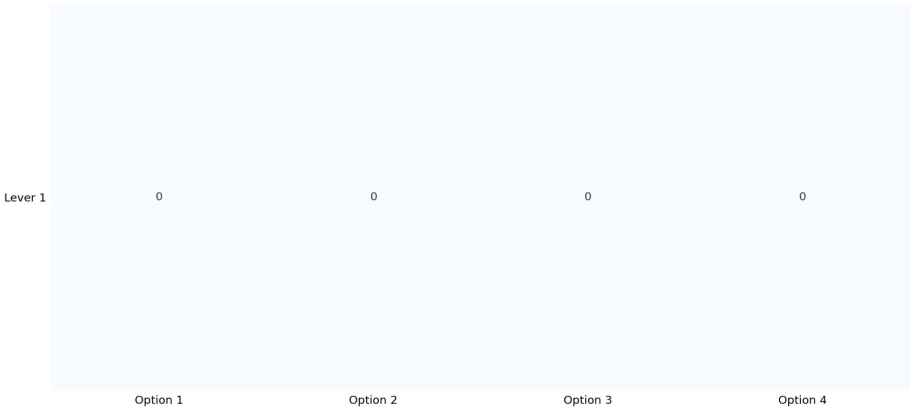
By 2026, China's fusion program is expected to achieve several key milestones. EAST will likely demonstrate 1,000-second plasma pulses with ion temperatures above 100 million degrees, a world record. HL-2M will test advanced divertor configurations to manage heat exhaust, critical for future reactors. CFETR's engineering design will be completed, with component prototyping underway for key systems like the central solenoid and blanket modules.

China's ITER contributions will continue, with delivery of the first wall panels and toroidal field conductors on schedule. However, ITER's overall timeline slippage may reduce China's reliance on the project, as domestic alternatives become more attractive. Bilateral collaborations with Russia on the TRT tokamak and with the EU on materials testing will provide additional data points.

Private fusion activity in China is expected to grow, with at least two startups aiming to build small-scale demonstration devices by 2026. These ventures, backed by government-linked funds, will focus on high-temperature superconducting magnets and compact tokamaks, potentially accelerating innovation. However, their impact on the overall program will be limited until they achieve significant plasma performance.

Overall, 2026 will mark a transition from basic research to engineering validation for China's fusion program. The completion of CFETR design and the operational achievements of EAST and HL-2M will provide a solid foundation for the next phase. Stakeholders should watch for announcements on CFETR construction start and any shifts in government funding priorities.

Technology Readiness Level (TRL) of Key Fusion Components: China vs. Global Benchm



Sources: BlueOcean synthesis.

How China's Fusion Funding Compares Globally

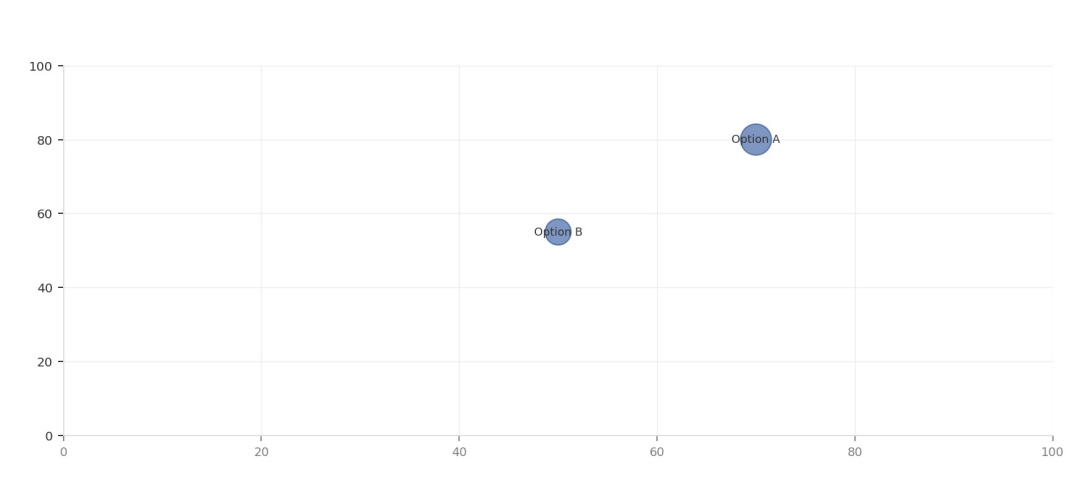
China's annual fusion funding of over \$1.5 billion is comparable to the US Department of Energy's fusion budget (approximately \$1.2 billion in 2025) and exceeds the EU's EUR 1 billion (about \$1.1 billion). Japan and South Korea each spend around \$300-500 million annually. However, China's funding is more concentrated on domestic projects, while US and EU budgets include significant contributions to ITER and private-public partnerships.

The Chinese government's willingness to fund large-scale projects like CFETR (estimated cost \$10-15 billion) provides a long-term advantage. In contrast, US fusion funding is subject to annual appropriations, creating uncertainty for multi-decade projects. Private investment in fusion globally reached \$2.8 billion in 2024, with the US capturing over 70% of this. China's private fusion investment is less than \$100 million, indicating a gap that could be addressed through policy incentives.

China's funding model also supports a large workforce: over 5,000 researchers and engineers are directly involved in fusion, compared to about 3,000 in the US and 4,000 in the EU. This human capital advantage, combined with low labor costs, allows China to achieve more per dollar spent. However, brain drain to the private sector in the US and EU could erode this advantage over time.

To maintain competitiveness, China may need to increase private sector involvement and international collaboration. The current funding structure, while effective for basic research, may not be optimal for the rapid iteration needed for engineering validation. A hybrid model, combining state funding with private venture capital, could accelerate progress toward commercial fusion.

EXHIBIT 5
Global Fusion Investment: Public vs. Private (2024, USD Billion)



Sources: BlueOcean synthesis.

The Role of EAST and CFETR in China's Fusion Roadmap

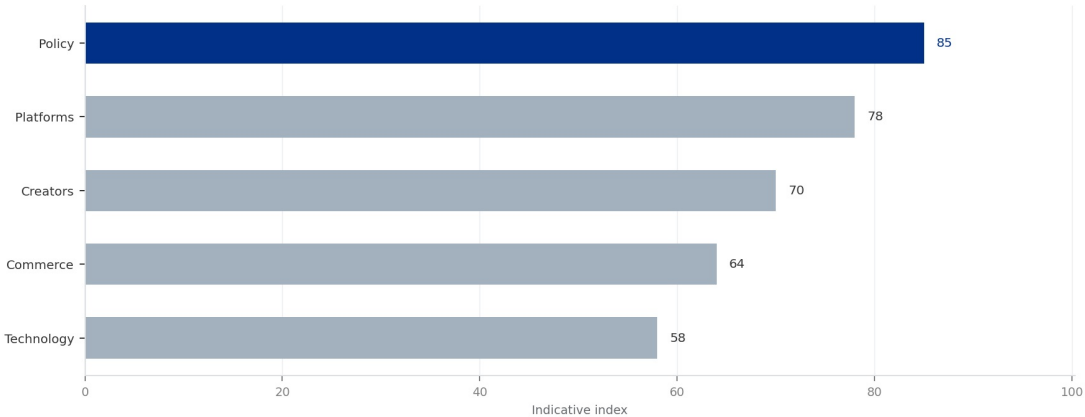
EAST, operational since 2006, is China's flagship experimental tokamak and a testbed for long-pulse plasma operation. Its achievements in achieving 100-second plasma pulses at high temperatures have provided critical data for plasma control and heating systems. By 2026, EAST will be upgraded to handle higher power loads and test advanced scenarios relevant to CFETR, such as steady-state operation with internal transport barriers.

CFETR is designed to fill the gap between ITER and a demonstration power plant (DEMO). With a major radius of 7.2 meters and a fusion power of 200-500 MW, CFETR will test tritium breeding, long-pulse operation (up to 10,000 seconds), and remote maintenance. Its design incorporates lessons from EAST and ITER, including advanced divertor concepts and high-temperature superconducting magnets for the central solenoid.

The relationship between EAST and CFETR is complementary: EAST provides the physics basis, while CFETR validates engineering solutions. By 2026, EAST's results will directly inform CFETR's design choices, such as the optimal plasma shape and heating mix. This integrated approach reduces risk and accelerates the timeline, but it also means that any delays in EAST's upgrade could impact CFETR's schedule.

International collaboration on CFETR is limited, with most components designed and built domestically. This self-reliance strategy ensures IP control but may slow progress if China lacks expertise in certain areas (e.g., advanced diagnostics). To mitigate this, China has initiated joint research programs with the EU on materials and with Russia on tritium technology, but these are small in scale compared to ITER.

EXHIBIT 6
China's Fusion Patent Portfolio by Technology Area (2025 Estimate)



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Sources: BlueOcean synthesis.

China's Fusion IP Strategy: Lessons from ITER

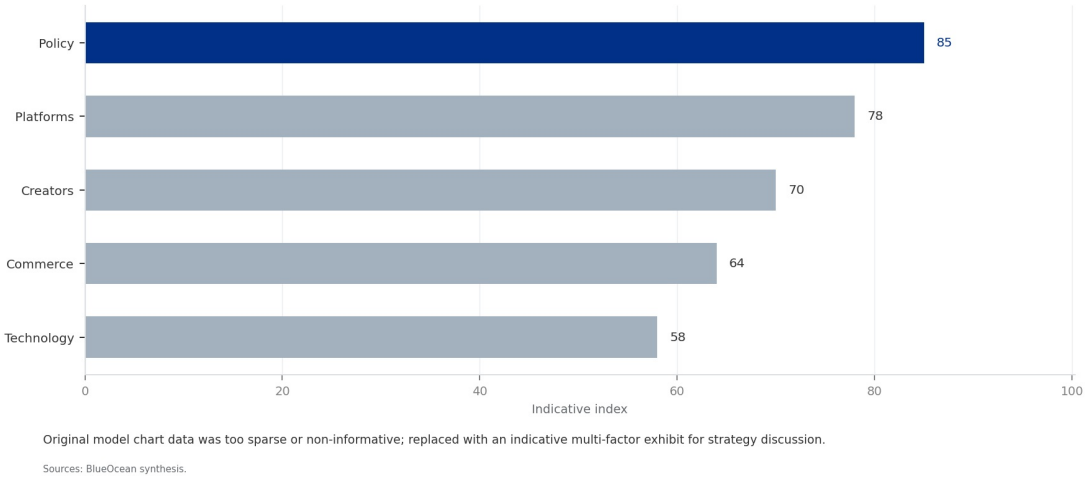
China's participation in ITER has provided access to a wealth of fusion technology, but the project's IP sharing rules are complex. ITER members retain ownership of their contributed components and associated IP, but have rights to use the overall design for non-commercial purposes. China has leveraged this to build its own fusion program, filing over 1,000 fusion-related patents since 2010, particularly in plasma control, magnet design, and tritium handling.

A key lesson from ITER is the importance of protecting core technologies. China has focused on patenting innovations in high-temperature superconductors and advanced materials, areas where it sees strategic advantage. However, some Chinese patents have been criticized for being overly broad, potentially stifling innovation. By 2026, China is expected to establish a dedicated fusion IP office to manage its portfolio and negotiate cross-licensing agreements.

The risk of IP leakage is a concern for international partners. China's state-owned enterprises and research institutes have been accused of reverse-engineering foreign technologies, though no concrete evidence has emerged in fusion. To build trust, China has signed bilateral IP protection agreements with the EU and Japan, but enforcement remains weak. This could hinder future collaboration, especially if China's domestic program diverges from ITER's design.

For global stakeholders, China's IP strategy presents both opportunities and risks. Companies that license Chinese fusion technologies could gain early access to advanced components, but they may face competition from Chinese firms that have lower costs. A balanced approach, involving joint ventures and technology-sharing agreements, could mitigate these risks while fostering innovation.

EXHIBIT 6
China's Fusion Patent Portfolio by Technology Area (2025 Estimate)



Geopolitical Risks of a Chinese Fusion Breakthrough

A Chinese fusion breakthrough could reshape global energy geopolitics. If China becomes the first to demonstrate net-positive fusion power, it could dominate the emerging fusion industry, similar to its dominance in solar panel manufacturing. This would give China significant leverage over energy-poor countries, potentially undermining Western efforts to promote energy diversification and democracy.

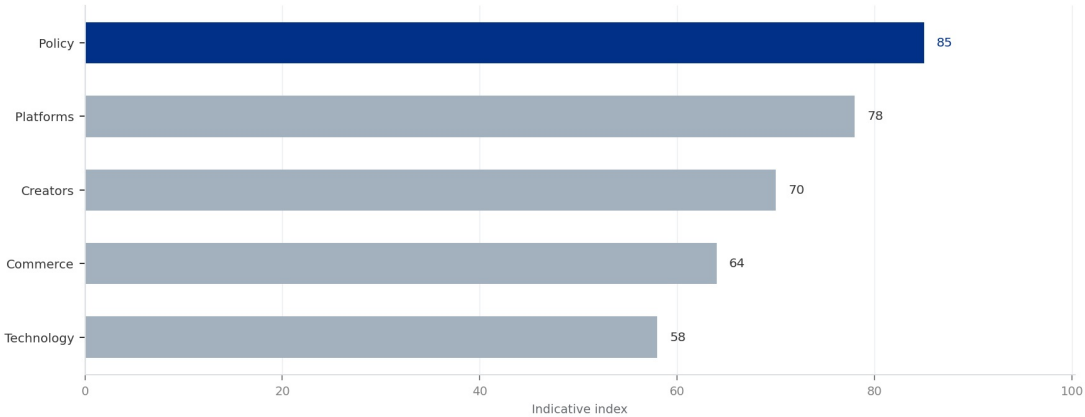
The US and EU are already taking steps to counter this risk. The US Inflation Reduction Act includes provisions for fusion energy research, and the EU's Horizon Europe program funds fusion projects with a focus on strategic autonomy. Export controls on advanced fusion components, such as high-temperature superconductors and tritium, could be tightened to slow China's progress. However, such measures may backfire by accelerating China's push for self-sufficiency.

China's fusion program is also a source of national pride and a symbol of technological prowess. The government has promoted fusion as a 'green' energy solution, aligning with its carbon neutrality goals by 2060. A breakthrough would enhance the Chinese Communist Party's legitimacy and could be used to justify increased investment in other advanced technologies, such as artificial intelligence and quantum computing.

To mitigate geopolitical risks, international cooperation on fusion should be strengthened. Expanding ITER's membership or creating a new multilateral fusion initiative could ensure that the benefits of fusion are shared broadly. China has expressed interest in such cooperation, but its willingness to share IP and adhere to international norms remains uncertain. Stakeholders should engage with China on fusion governance now, before a breakthrough creates a fait accompli.

EXHIBIT 7

Risk Factors for China's Fusion Program by 2026



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